Maturity and Measurement Together: Honest AI4SE Assessment Beyond the Multiplier Claim

Juan Qiu*, Qilong Yan*
*Inspire Group, China
Emails: juan.qiu@inspiregroup.com, qlyan@inspiregroup.com

***Abstract*—Enterprise sponsors of AI-for-software-engineering (AI4SE) programs often ask for a single developer-productivity multiplier, yet slogans rarely separate belief from measured delivery. Multi-dimensional guidance such as SPACE and DORA already warns against collapsing activity, efficiency, and system delivery into one number. Honest AI4SE assessment should *pair capability maturity with labeled delivery measurement*; a multiplier claim is not a substitute for that pairing. We propose a dual-track assessment protocol and illustrate it in one industrial experience at Org-A (∼6-week enablement pilot). We do not claim a completed design-science evaluation or an action-research intervention. Track 1 is facilitator-guided capability maturity before/after (`process_maturity`). Track 2 is delivery-result metrics—Throughput and cycle time (CT)—under a fixed Product Backlog, team, and iteration timebox, elicited here as `metric_estimate`. A third label, `metric_platform`, is reserved for the same Throughput/CT metrics computed later from Org-A's *requirements management platform* (Product/Sprint Backlog workflow; Git only where commit timestamps are needed) during real iterations; this paper does not yet report those platform rows. Honest assessment here means every published cell carries a signal-type label, and Throughput/CT medians stay promissory on an estimate→measured *promotion path* until matching `metric_platform` values validate or refute them. At Org-A, both delivery-metric Before and After were expert estimates (estimate-to-estimate): maturity lifts were modest ($+0.33$ to $+1$ on a 1–5 mean), while six pairs' median $\sum$ story-point volume improve Δ was about $+100\%$ in-window and CT1 speed-improve median was ≈ 1 (about $2\times$ faster P75 code cycle time under the same promissory label). Those figures are estimate-to-estimate expectations, not causal effect sizes or platform-validated ROI. Full six-decimal medians appear only in tables and the public CSV pack.**

***Index Terms*—AI for software engineering, maturity assessment, developer productivity measurement, SPACE, DORA, industrial experience report**

I. Introduction

Industry narratives about AI coding assistance often collapse value into a single multiplier ("$N\times$ faster," "thousands of percent improvement," or similarly non-comparable slogans). Multi-dimensional views of developer productivity caution against that collapse [1]: Satisfaction, Performance, Activity, Communication/collaboration, and Efficiency/flow can move in different directions, so a single throughput slogan is a poor decision aid. Two further context failures make such slogans unsafe as management evidence. First, a large gain confined to the *build* (code-generation) stage is not the same as an equal gain across the end-to-end lifecycle: surrounding stages can remain near human speed, so cycle-time reclaimed is smaller than a build-only ratio suggests—Anthropic's AI-native SDLC playbook likewise argues that when build accelerates, plan, review/test, and deploy can stay human-speed bottlenecks [2]. Second, any $N\times$ claim needs an explicit comparator and fixed measurement conditions; without them the number is not interpretable. Published experimental evidence typically reports *moderate, conditional* gains—for example, roughly 26% more completed tasks across three pooled *field* experiments [3], and about 21% shorter time (wide CI) on a complex enterprise-grade task in a Google-engineer RCT that the authors describe as a lab study [4]—while a controlled coding-task experiment reported about 55.8% faster completion with an AI pair programmer [5]. DORA research popularizes system-level delivery metrics (lead time for changes, deployment frequency, change fail rate, time to restore) [6], which are distinct from story-level throughput and cycle time. Perception can diverge from measured effect: experienced open-source developers in one RCT of early-2025 AI tools forecasted faster completion yet were about 19% slower when AI was allowed [7]. Sponsors still ask for a defensible answer to "how much did we improve?" under disclosure and instrumentation constraints.

What they still lack is a *management-usable assessment protocol* that (i) pairs capability maturity with delivery-outcome *signals*, (ii) labels how each number was obtained (estimate versus measured), and (iii) refuses to treat a multiplier claim or workshop consensus as platform-measured ROI. Single-metric "same work, faster" framings break down when AI4SE changes how work is decomposed, verified, and shipped. Story-point volume, cycle-time endpoints, and maturity rubrics must therefore be held fixed *by protocol* if Before/After comparisons are to mean anything; otherwise teams re-point work, change iteration timeboxes, or conflate coding activity with end-to-end delivery.

We address two practice questions illustrated in one industrial engagement. These questions are practice questions for an experience report, not research-track empirical findings or method-construction prompts alone:

- **RQ1.** How can AI4SE enablement improvement be as-

J. Qiu is Principal AI4SE Consultant at Inspire Group (Ph.D., Software Engineering, Tongji University; 18+ years in software engineering, R&D productivity, and enterprise architecture). Q. Yan is AI4SE Engineering Advisor at Inspire Group (M.S., Computer Science, Sichuan University; 10+ years in enterprise architecture, DevOps, and AI-agent engineering). Inspire Group is a technology-services firm formed from the former Thoughtworks China business after its acquisition by Hillhouse Capital (announced 2025).

sessed jointly from delivery-result metrics (e.g., Throughput and cycle time) and from process-capability maturity in an industrial pilot?
- **RQ2.** How can such dual-track assessment be reported honestly when delivery-result metrics are still expert estimates (`metric_estimate`) rather than platform-measured values (`metric_platform`)?

Methodological stance. This manuscript is an *industrial experience report*: a single-case illustration of a dual-track maturity-and-measurement assessment protocol, reported with attention to context, chain of evidence, and threats [8], under client disclosure constraints. The protocol itself is a designed artifact in the sense of design-science research [9], but we do not claim a complete design-science evaluation cycle, multi-case theoretical saturation under case-study guidelines [8], or an action-research intervention with researcher-owned change agency. We report what Org-A's pilot made observable under the protocol, including modest maturity movement and promissory workshop Δs, and we foreground threats where evidence is thin.

Contributions. This paper contributes: (1) a diagnosis of why a bare multiplier claim fails as management evidence under AI4SE (build-local speedup confused with end-to-end gain; missing comparator and fixed conditions; missing signal-type labels); (2) a dual-track assessment protocol that uses maturity and measurement *together*: capability maturity (`process_maturity`) plus Throughput/Cycle-Time signals with a three-way `signal_type` taxonomy (two instantiated at Org-A; `metric_platform` reserved) and three design anchors (comparability, separability, honesty); (3) a single-case illustration at Org-A ($\sim$6-week enablement pilot) with every table number frozen in a public derived-data pack; and (4) managerial implications: a dual-track dashboard, an estimate$\rightarrow$measured promotion path, explicit falsifiers, and threats to validity. We do not claim a new RCT effect size, a scale-out ROI, a completed platform-measured closed loop at Org-A, or a universal maturity-model theory. The transferable artifact is the assessment discipline (maturity with labeled measurement); After-condition productivity rows in this case are workshop elicitations, not measured delivery histories. Where this paper uses stage names, *Explore* is the short enablement window that produced the dual-track cells below; *Consolidation* and *Scale* are later adoption phases discussed only as measurement boundaries (not as additional Results tables).

Positioning. Relative to SPACE [1], DORA [6], and AI-assisted SE maturity models [10]–[12], our niche is a consulting-embedded dual-track *assessment* protocol that jointly publishes maturity before/after and stratified delivery-metric signals under disclosure constraints, with an estimate-vs-measured promotion path and a public derived pack—not a replacement for those maturity frameworks and not a multi-site causal evaluation.

II. Industrial Context

Org-A is a large global container-shipping and logistics enterprise with a substantial, multi-site in-house IT / software-engineering organization that builds and operates enterprise-scale applications supporting core commercial, customer-facing, and operational processes. Software delivery follows a multi-team agile model, with Product Backlog Items (PBIs) managed in a shared requirements management platform and a conventional Dev/BA/QA role split. It ran a short, approximately six-week AI4SE enablement pilot—a phase-one explore / co-working engagement with a consulting facilitation team, not a scale-out program. Delivery work on a pilot vehicle served as a *validation context* for method enablement and evidence collection; phase success was not defined as "scale-ready." Baseline tooling included conventional IDE-centric development with emerging AI assistants; the pilot introduced a spec-driven method pack (playbook, executable recipes, and facilitation rituals) rather than a single vendor coding agent alone. Stakeholders for the executive readout included a sponsor circle (engineering leadership) and delivery leads from the pilot team; they repeatedly asked for productivity proof suitable for funding and phase-gate discussion.

Quality as a precondition, not a missing track. At Org-A, product quality was already treated as a standing red line in sponsor discussions: AI4SE enablement was framed as accelerating delivery *efficiency under non-degrading quality*, not as trading quality for speed. The engagement therefore did not commission a parallel quality-instrumentation workstream for this paper; facilitators defaulted to Org-A's existing quality gates and indicators (e.g., defect escape rate and defect density under the client's established definitions) rather than inventing a new quality scoreboard beside Throughput/CT. Those quality indicators are acknowledged as governance constraints here; this manuscript does not re-analyze or claim measured quality deltas. **Non-claim:** no quality before/after Δ is reported; non-degradation is a governance stance, not a measured finding of this paper.

The engagement's relevant outputs for this paper were: (i) a facilitator-guided maturity re-assessment before and after the enablement window; (ii) a facilitated effectiveness workshop under fixed Product Backlog (identical PBIs), team size, and iteration window (timebox); and (iii) an executive readout that stated signal-type boundaries to management. Those questions are legitimate; answering them with a single invented multiplier is not. The consulting response was a dual-track maturity-and-measurement assessment, not a single-multiplier readout. Workshop expert estimates were positioned as promissory expectations under an After prompt framed as high-consensus SDD adoption, not as platform-measured ROI. Modest maturity movement over a few weeks was treated as an expected finding, not as a narrative failure. Explicit non-claims were recorded for the phase: phase success $\neq$ scale-ready; `metric_estimate` Throughput/CT results $\neq$ `metric_platform` ROI.

Why a dual track was required. Sponsors ask whether

TABLE I
SIGNAL-TYPE TAXONOMY USED IN REPORTING.

signal_type	Meaning in this paper
process_maturity	Facilitated process-capability maturity assessment (domain/capability levels)
metric_estimate	Delivery-result metrics (Throughput/CT) from facilitated SME / pair judgment—**both** Before and After rows at Org-A
metric_platform	Same Throughput/CT metrics from the requirements management platform (Product/Sprint Backlog workflow; Git if needed for CT1)—**not yet reported** for Org-A Before/After in this paper

TABLE II
THREE DESIGN ANCHORS FOR THE PRODUCTIVITY WORKSHOP.

Anchor	Operational implication
Comparability	Hold Product Backlog (PBI list), team size, and iteration window (timebox) constant so Before→After differences are not explained by those three condition drifts (this does *not* prove causal attribution to the method alone)
Separability	Freeze story points after joint estimation; After must not re-point
Honesty	Report high-consensus expert judgment as promissory expectation, not platform-measured ROI

AI4SE enablement improved delivery efficiency. That ask has two complementary faces: *delivery-result metrics* (e.g., Throughput and cycle time) make the magnitude of expected or measured lift visible; *process-capability maturity* shows whether efficiency-related capabilities are being institutionalized. Reporting only maturity under-answers "how much throughput / cycle time?"; reporting only metrics ignores institutionalization. Publishing both under distinct `signal_type` labels—`process_maturity` for capability assessment, and `metric_estimate` / `metric_platform` for the same Throughput/CT definitions—keeps the two faces on separate scales and blocks collapsing them into one unvalidated multiplier.

For anonymity, the client is labeled Org-A only; delivery-metric elicitation groups appear as Pair-1…Pair-6; no participant or facilitator personal names appear in this paper or in the accompanying public-data pack. Identifiable product features from the pilot vehicle are also withheld.

III. MEASUREMENT SCHEME

The scheme answers two management questions on one dashboard: *Are capabilities moving?* and *What throughput / cycle-time improvement can we expect under high consensus?* Every published row carries a `signal_type` (Table I). Every protocol choice traces to one of three anchors (Table II).

In this paper, *requirements management platform* means Org-A's shared system for Product Backlog / Sprint Backlog items and their workflow states (optionally linked to Git for commit timestamps)—not a full Application Lifecycle Management suite claim, and not a named vendor product. The workshop's fixed work list is the Product Backlog (identical PBIs for all pairs), hosted on that platform. Reader cue: `process_maturity`=facilitated maturity (not delivery ROI); `metric_estimate`=promissory Throughput/CT judgment (Before *and* After); `metric_platform`=same metrics from the platform in real iterations (**not reported** here).

Critical boundary: Before is also an estimate. Org-A is a mature agile organization with platform history, yet the workshop did *not* bind Before to a historical platform window. Both Before (traditional delivery) and After (spec-driven method under an After prompt framed as high-consensus SDD adoption) were elicited as `metric_estimate` on the *same* fixed Product Backlog, team, and iteration timebox, so that the only intended contrast is method under those controls. Historical platform throughput on past sprints typically does not map onto that exact PBI set and timebox; After under the new method also had no closed platform history yet. Consequently, Before→After Δ is an *estimate-to-estimate* contrast (a promissory method lift)—not measured-baseline→estimate, and not platform→platform. We treat that as a first-order credibility issue (Section VI): the gap can reflect shared optimism ("imagination gap") as well as genuine method expectation, which is why `metric_platform` validation remains mandatory before ROI language.

A. Capability Maturity Lens

Capability maturity is assessed with a facilitator-guided instrument covering **6 domains** $\times$ **18 capabilities** (3 per domain) on levels **L1–L5** (Initial $\rightarrow$ Toolized $\rightarrow$ Engineered $\rightarrow$ Systematized $\rightarrow$ Autonomous). English domain labels used in derived data are: Efficiency Foundation; Process Layer; Engineering Methods & Discipline; Tooling Layer; Human–Agent Collaboration; and Organization & Culture. Example capabilities include value goals and outcome hypotheses (D1.1), AI4SE workflow design (D2.1), spec-driven expression (D3.1), agent runtime and permissions (D4.1), human-in-/on-/out-of-the-loop risk routing (D5.2), and practice champions and training (D6.1); the full 18-item list ships in `maturity_capabilities.csv`.

Scoring packaging. A consulting facilitator walks sponsors and delivery leads through each capability against short L1–L5 anchors (observable practice cues), records a consensus level, and revisits the same instrument after the enablement window. At Org-A, Before and After scores were collected by the same facilitation team around the start and end of the $\sim$6-week pilot; the effectiveness workshop was a separate session under fixed conditions and is *not* the maturity instrument. The public pack includes a concise scoring guide (`maturity-scoring-guide.md`) with level anchors and facilitation notes; it is an industrial rubric for repeated use, not a psychometrically validated scale. No inter-rater reliability study is claimed here: scores are expert-facilitated consensus under time pressure.

Ordinal caveat. Domain means in tables are convenience averages of three ordinal levels on a 1–5 encoding; they support before/after narrative comparison within this instrument, not interval-scale inference or cross-vendor maturity rankings.

TABLE III
CYCLE TIME (CT) VARIANTS USED IN THE WORKSHOP.

CT	Meaning	Endpoints (start → end)
CT1	Code ($\mathrm{CT}_{\mathrm{code}}$)	DEV accept → latest related commit
CT2	Delivery ($\mathrm{CT}_{\mathrm{delivery}}$)	DEV accept → QA testing completed

DEV accept = first Product Backlog → Required Analysis transition on the story. CT∗ durations are story-grain; pair summaries use P75.

A $+0.33$ domain mean typically reflects a one-level lift on a single capability with the other two unchanged.

Scores are labeled `process_maturity`: they track organizational capability change, not product quality or business outcome, and they are *not* platform-measured. The instrument is intended for repeated assessment across transformation phases; a short pilot should be expected to move only a subset of capabilities by one level. Relative to emerging AI-assisted SE maturity models—organizational governance for AI-assisted development [10], individual/organizational competency measurement [11], and codebase feedback-loop progression toward autonomy [12]—this lens is a *program-local* 6×18 checklist for dual-track reporting, not a replacement org-wide maturity theory.

B. Productivity Signal Lens

Throughput. In the metrics specification, $\mathrm{Throughput} = \sum(\mathrm{SP\ of\ completed\ stories\ in\ window})/\mathrm{PersonDays}$ (points / person-day). Here *window* denotes the fixed agile *iteration timebox* used by the workshop (not the multi-week enablement pilot). Completion is the first transition to QA testing completed (or platform-equivalent). Under the workshop's fixed team $\times$ iteration window, PersonDays is approximately constant, so $\sum$ **SP** in the window is the primary floor report and is monotone-equivalent for Before/After Δ.

Cycle time (CT). We write **CT** for *cycle time*: the elapsed duration of a story between two defined platform workflow endpoints (not the iteration timebox, and not DORA lead time for changes). The workshop primary-reports two story-level CT variants (Table III):

- **CT1** (*code* cycle time; spec alias $\mathrm{CT}_{\mathrm{code}}$): from DEV accept to the latest related commit—coding-centric elapsed time.
- **CT2** (*delivery* cycle time; spec alias $\mathrm{CT}_{\mathrm{delivery}}$): from the same DEV-accept start to QA testing completed—includes handoff into test-done.

$\mathrm{CT}_{\mathrm{prod}}$ (to production release) is defined in the metrics spec but not primary-reported on the floor (non-dev release noise). Within-pair CT summaries use **P75** at story grain. After (spec-driven) work may be planned as Changes, but statistics fold back to stories before P75.

Paired improve Δ. Throughput improve is $(\mathrm{TP}_{\mathrm{after}} - \mathrm{TP}_{\mathrm{before}})/\mathrm{TP}_{\mathrm{before}}$. CT speed improve (primary narrative) is $(\mathrm{CT}_{\mathrm{before}} - \mathrm{CT}_{\mathrm{after}})/\mathrm{CT}_{\mathrm{after}} = \mathrm{speedup} - 1$ for the chosen CT variant (CT1 or CT2); we also report speedup $\mathrm{CT}_{\mathrm{before}}/\mathrm{CT}_{\mathrm{after}}$ and duration reduction $(\mathrm{CT}_{\mathrm{before}} - \mathrm{CT}_{\mathrm{after}})/\mathrm{CT}_{\mathrm{before}}$, without conflating denominators. Cross-pair primary claims use the **median and IQR** of the six pair Δs—not the mean of means, and not the maximum pair.

TABLE IV
FIXED WORKSHOP CONDITIONS (COMPARABILITY).

Condition	Workshop setting
Product Backlog	Identical PBI list for all pairs (platform-hosted backlog; SP frozen after Stage 1)
Team size	Fixed roster (study example: 5 Dev + shared BA/QA fractions)
Iteration window	Agile iteration timebox: 2 weeks / 10 working days
Story points	Frozen after Stage-1 joint estimation; After must not re-point

Workshop protocol (fixed conditions). Table IV summarizes the fixed conditions. Twelve participants form **six independent pairs** (Pair-1…Pair-6) that estimate Before (traditional delivery) vs. After (spec-driven method pack at high consensus) without mid-workshop sharing, to limit anchoring. Stages: (1) clarify Product Backlog PBIs & freeze SP (read-only thereafter); (2) Before estimates of which stories finish in the iteration window and story-level CT1/CT2; (3) After estimates (scope as Changes, map back to stories, sum *frozen* SP, recompute P75 CT); facilitator aggregates six pair Δs into median / IQR (and optionally min/max). The protocol forbids averaging all Before then all After into one global Δ, and forbids dropping pairs for "ugly" numbers except for documented protocol violations. All effectiveness rows are `metric_estimate`.

C. Combined Management Dashboard

Table V states the joint reading rule. Track both lenses with explicit `signal_type` labels; never substitute estimate for measured ROI; never treat flat maturity as "no value" when method assets and promissory expectations still exist. Operationally, the dashboard is a *two-track* artifact: one track is capability levels over time; the other is stratified productivity signals with the estimate→measured promotion path reserved for `metric_platform` rows that do not yet exist for Org-A's After condition. The closed loop for later phases is: publish workshop medians as expectations, then validate or refute them with `metric_platform` telemetry under the same metric definitions. A third anti-pattern to avoid is mixing denominators—reporting duration reduction % as if it were speed-improve %—which the protocol tables explicitly separate.

Coverage vs. SPACE and DORA (deliberate omissions). SPACE warns that Satisfaction, Performance, Activity, Communication/collaboration, and Efficiency/flow can diverge [1]; this scheme operationalizes primarily Efficiency/flow (Throughput, CT) plus a process-capability track, and does *not* claim DevEx satisfaction surveys, communication-network metrics, or a new quality/defect Per-

TABLE V
DUAL-LENS MANAGEMENT DASHBOARD (READING RULES).

Lens	Honest read
Maturity	Modest short-pilot movement is an expected finding, not automatic failure
Productivity signals	Workshop = promissory expectation under fixed conditions; close the loop later with `metric_platform`
Joint use	Capability lift $\neq$ ROI; estimate $\neq$ platform-measured; keep both tracks labeled

formance instrument as primary Org-A rows. That omission is not a claim that quality is unimportant: Org-A sponsors treated quality as a default, non-negotiable constraint (Section II), and AI4SE efficiency asks were posed under "quality must not degrade." Accordingly, the pilot retained the client's existing quality requirements (illustratively defect escape rate and defect density) instead of adding a separate workshop quality lens; this paper simply does not expand those retained indicators. DORA-style system metrics (deployment frequency, lead time for changes, change fail rate, time to restore) [6] likewise remain out of the workshop primary report: CT1/CT2 are story-grain delivery intervals under a fixed Product Backlog and iteration timebox, not org-level change lead time or reliability. Sponsors who need SPACE satisfaction, DORA reliability, or a published quality before/after pack must instrument those tracks separately; omitting them here is intentional under disclosure, cadence, and the client's "quality already governed" stance—not a claim that Throughput/CT alone suffice for all management decisions.

Sponsor decision tree (anti-misquote). When maturity Δ is modest and workshop medians are large—the Org-A pattern—sponsors should: (1) fund next-phase method enablement and instrumentation, not declare ROI; (2) schedule `metric_platform` validation against the same definitions before any funding slide quotes $+102\%$ $\sum$ SP or $\approx 2\times$ CT speedup as achieved productivity; (3) forbid retelling Pair-3 maxima as the organizational headline when the primary claim is median+IQR; (4) ask who owns the dual dashboard (engineering leadership + delivery leads jointly) and what incentive pressure would punish honest estimate$\neq$measured divergence; (5) refuse to narrate a build-stage or workshop $N\times$ as end-to-end lifecycle ROI without an explicit comparator and fixed conditions. Executive-facing narrative may round to "about $+100\%$ $\sum$ SP under workshop conditions (promissory)" while precision stays in CSV/tables.

Falsifiers. The *scheme* fails if published rows drop `signal_type`, if After re-points work, if workshop maxima replace median+IQR as primary claims, or if estimate rows are promoted to ROI without platform-measured confrontation. The *enablement* fails if later `metric_platform` Δs remain near zero under the same definitions while method consensus was claimed high, if maturity domains tied to the method pack stay flat across multiple reassessment cycles with no asset adoption evidence, or if efficiency narratives are advanced while Org-A's quality red-line indicators (e.g., defect escape / density under client definitions) are knowingly breached or dropped from governance. Either failure is reportable news—not a reason to invent a multiplier.

Ethics and politics of the dashboard. Throughput/CT estimates can become weapons in headcount or vendor debates; facilitators must keep ownership of labels with named sponsors, surface risk/DevEx beside volume when available, and treat refusal to schedule platform-measured validation as a governance red flag rather than a data gap to paper over.

Application note (executive use). On one slide, report maturity domain means and workshop median+IQR, each with `signal_type`; use maturity for "are we building lasting capability?" and workshop signals for "what delivery lift should we expect under high consensus?" Do not collapse the two into one multiplier; do not headline Pair-3 maxima when the primary claim is median+IQR; apply the decision tree above when maturity is modest and estimates are large. Research implications of this dual-track reading appear in Section V.

IV. CASE APPLICATION

We apply the scheme to Org-A's $\sim$6-week pilot. All quantitative cells below are copied from `public-data/derived/*.csv`.

A. Process Narrative

Phase one followed a short enablement cadence rather than a delivery-acceptance program. Sponsors opened with a productivity-proof ask under disclosure limits; facilitators answered with dual-track maturity-and-measurement assessment instead of a single multiplier because Org-A already had quality red lines and an immature instrumentation path for After under the new method.

Why these protocol choices. We fixed Product Backlog, team, and iteration timebox so Before/After workshop contrasts would not be explained by scope or calendar drift; we froze story points after joint estimation so After could not "win" by re-pointing; and we deliberately elicited *both* poles as `metric_estimate` on that same PBI list because a historical platform window would not map onto the fixed backlog used for After SDD scoping. Those choices trade measured-baseline credibility for within-workshop comparability—a trade we surface rather than hide (Section III).

What the weeks looked like. A kickoff week locked diagnosis, sponsor questions, and a pilot vehicle as the validation context for method assets. Four workshop weeks then co-created a spec-driven method pack (playbook, opportunity specs, executable recipes/skills) with the pilot team on real PBIs. Mid-engagement, facilitators tightened wording for executive materials: phase success meant reusable method assets and bounded evidence—not "all business scope done," not scale-ready, and not estimate-as-ROI. Near the start and end of the enablement window, the same facilitation team ran the 6×18 maturity re-assessment (`process_maturity`); learning-to-the-test was possible and was *not* blinded in this

TABLE VI
CAPABILITY-LEVEL LIFTS WITH $\Delta \neq 0$ (EXCERPT). ALL ROWS: PROCESS_MATURITY. SOURCE: MATURITY_CAPABILITIES.CSV.

ID	Capability (short)	Before→After	Δ
D1.3	Efficiency & Flow Measurement	L1→L2	1
D2.1	AI4SE Workflow Design	L2→L3	1
D2.2	Small-Batch Task Decomposition	L1→L2	1
D3.1	Spec-Driven Expression	L2→L3	1
D3.2	Context Organization	L1→L2	1
D4.1	Agent Runtime & Permissions	L2→L3	1
D5.2	HITL/HOTL/HOOL Risk Routing	L1→L2	1
D6.1–3	Champions / Community / Adoption	L2→L3	1 each

pilot. Separately, the effectiveness workshop (six pairs, independent estimates) produced Table VIII's promissory Δs. A closing executive readout presented both lenses with signal-type labels, retained quality gates, and a next-step path of sedimentation before any scale-out discussion.

What we hit and did not do. The main friction was executive appetite for a single $N\times$ number while instrumentation for `metric_platform` After was not yet closed; the honest move was to publish estimate-to-estimate contrasts with hard labels rather than invent platform ROI. We did *not* bind workshop Before to a matched historical platform window on the same PBI set; close After with `metric_platform` telemetry; or publish a parallel quality before/after pack beyond existing client gates.

B. Maturity Before/After

Table VII and Fig. 1 report domain means (`process_maturity`). **Overall deltas are modest:** domain lifts range from $+0.33$ to $+1.00$; most domains remain in the low-L2 / low-L3 mean band after re-assessment. Largest lifts: Organization & Culture (D6, $\Delta = 1$); Process Layer and Engineering Methods & Discipline (D2, D3, $\Delta = 0.66$ each). Smaller lifts: Efficiency Foundation, Tooling Layer, and Human–Agent Collaboration ($+0.33$ each).

Capability detail (18 items in `maturity_capabilities.csv`) shows several capabilities remaining at L2 with $\Delta = 0$ (e.g., D1.1, D1.2, D2.3, D3.3, D4.2, D4.3, D5.1, D5.3). Table VI excerpts capabilities with non-zero Δ to illustrate where movement concentrated. After the pilot, D1 and D5 still have zero capabilities at L3+. L1 shortfalls cleared in D1, D2, D3, and D5 (L1 counts $1 \to 0$); D6 after has three capabilities at L3+. Tooling and verification depth did not jump to systematized levels in this window.

Explicit finding: a short (∼6-week) enablement pilot ⇒ limited maturity movement is an *expected* finding, not a failure to report. Capability curves are slow relative to workshop promissory expectations; both lenses must stay on the dual dashboard without forcing a success narrative.

C. Workshop Effectiveness Signals

Under fixed Product Backlog / team / iteration window and frozen SP, six pairs produced Before/After estimates. **Interpretation label (required):** promissory expectation under an

TABLE VII
MATURITY DOMAIN MEANS BEFORE/AFTER (ORG-A). ALL ROWS: PROCESS_MATURITY. SOURCE: MATURITY_DOMAIN_SCORES.CSV.

ID	Domain	Before	After	Δ
D1	Efficiency Foundation	1.67	2	0.33
D2	Process Layer	1.67	2.33	0.66
D3	Eng. Methods & Discipline	1.67	2.33	0.66
D4	Tooling Layer	2	2.33	0.33
D5	Human–Agent Collaboration	1.67	2	0.33
D6	Organization & Culture	2	3	1

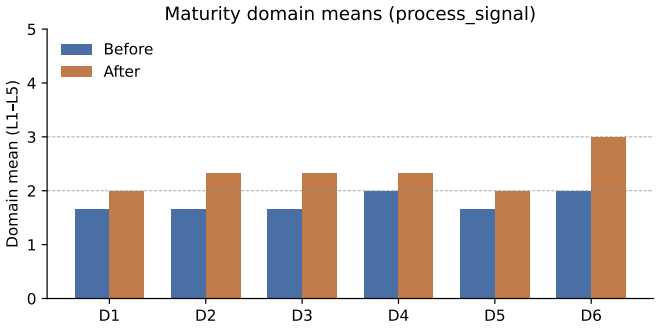


Fig. 1. Maturity domain means before/after (`process_maturity`). Horizontal guides at L2 and L3.

After prompt framed as high-consensus SDD adoption—**not** measured ROI, and **not** `metric_platform`. **Both poles are estimates:** Before (traditional delivery on the fixed PBI list) and After (spec-driven method under that After prompt) are `metric_estimate`; Table VIII is therefore an estimate-to-estimate contrast (Section III), not a platform baseline versus a workshop forecast.

Throughput (volume construct). Table VIII reports cross-pair aggregates for $\sum$ SP volume improve Δ in the iteration window: median 1.017544 (IQR 0.039384), range $[0.76,\ 1.472727]$ (source CSV; absolute Throughput rates $\sum$SP/PersonDays are not reported as primary cells). The median fractional improve of ≈ 1.02 corresponds to roughly $+100\%$ completed story-point volume in the fixed timebox as *estimated* by pairs—about a doubling of in-window $\sum$ SP under workshop conditions, not organization-wide ROI.

Cycle time (speed construct). Separately, CT1 speed-improve Δ at P75 has median 1 (IQR 0.1875); CT2 speed-improve Δ at P75 has median 1.018519 (IQR 0.616402). Speed-improve $\Delta = (\mathrm{CT}_b - \mathrm{CT}_a)/\mathrm{CT}_a$ equals speedup-1; a CT1 median of 1 therefore corresponds to roughly $2\times$ faster P75 code cycle time under the same promissory label. We do *not* claim that throughput "also" sped up $2\times$: $\sum$ SP improve Δ and CT speed-improve Δ are different constructs (volume change vs. duration speedup) and must not be narrated as one interchangeable multiplier. We do not cherry-pick the maximum pair for the primary claim.

Table IX summarizes pair-level throughput and CT P75. Pair Δs vary (e.g., Pair-5 throughput improve 0.76; Pair-3 1.472727), reinforcing median+IQR over max-pair story-

telling. Executive narrative may say "about $+100\%$ $\sum$ SP (volume) and about $2\times$ CT speedup (duration), both workshop estimates"; tables keep full precision and keep the two constructs separate.

D. Reading Both Lenses Together

Jointly, Org-A shows *modest* capability movement (`process_maturity`) alongside *magnitude-large but unvalidated* workshop promissory expectations (`metric_estimate`). That combination is coherent under the dashboard rules: capability lift $\neq$ ROI; estimate $\neq$ platform-measured. Published experiments on AI coding assistance typically report moderate conditional gains—for example, $\sim$26% completed-task increase across three pooled field experiments [3] and $\sim$21% shorter task time (wide CI) in an enterprise-engineer lab RCT [4]—while a controlled coding-task experiment reported about 55.8% faster completion [5]; perception can also diverge from measurement [7]. Our promissory workshop medians ($\approx$ $+102\%$ $\sum$ SP; CT speed-improve Δ $\approx$ 1) sit well above those published experimental point estimates; that gap is expected under different constructs and elicitation modes—high-consensus After prompting, frozen local Product Backlog, story-point volume vs. completed-task or wall-clock constructs, and optimistic SME judgment—and is *not* evidence that Org-A "beat" published trials. We do not equate workshop medians with RCT effect sizes, nor do we treat controlled-task speedups as portfolio ROI; both sit far from folklore $N\times$ organizational multipliers once labels are respected.

Three joint readings follow for sponsors: (1) **Timescale mismatch is expected.** Culture/process capabilities (D2, D3, D6) moved more than tooling/verification depth; workshop $\sum$ SP expectations moved faster than maturity means—do not force them onto one success narrative. (2) **Disagreement is informative.** Throughput IQR is tight (0.039384) while CT2 speed-improve IQR is wide (0.616402); pairs agree more on volume than on delivery-endpoint speed. (3) **Promotion path.** Keep workshop rows labeled `metric_estimate` until platform-measured rows exist; then compare, do not overwrite.

The managerial move is to keep both tracks visible and to schedule `metric_platform` validation rather than declaring victory from either lens alone.

V. Discussion

Observation for RQ1 (joint assessment from metrics and maturity). Org-A shows how the two faces work together: capability domain Δ ranged $+0.33$ to $+1$ on a 1–5 ordinal mean scale (`process_maturity`), while Throughput/CT delivery-result estimates reached median $\sum$ SP volume improve $\Delta = 1.017544$ and CT speed-improve $\approx$ 1 (`metric_estimate`)—modest process-capability movement beside magnitude-large promissory delivery metrics on their native scales. Culture/process domains (D2, D3, D6) moved more than tooling depth within six weeks; delivery-metric expectations still should not be forced onto one success score with maturity means. Relative to emerging AI4SE maturity models [10]–[12], our 6×18 checklist is a program-local process track that should be read *beside* labeled delivery-result metrics, not as a substitute ROI meter or as a claim to replace those broader maturity frameworks.

Observation for RQ2 (honest reporting under estimate-to-estimate metrics). Because delivery-metric Before and After rows are both `metric_estimate`, Table VIII cannot be retold as platform-proven lift (`metric_platform`). Honest dual-track reporting therefore: (i) keeps Throughput/CT medians as labeled expectations; (ii) instruments the same Throughput/CT1/CT2 definitions on requirements platform/Git; (iii) compares and revises method assets or expectations where they diverge; (iv) forbids silent label drops when platform-measured lifts are smaller. Closing that loop at Org-A remains future work; this paper's contribution is the reporting discipline that keeps the empty After `metric_platform` cell visible instead of hiding it.

Against $N\times$ myths. Literature point estimates ($\sim$21–26% in field/enterprise-task settings [3], [4] vs. larger controlled-task gains such as $\sim$55.8% [5]) already contradict folklore organizational multipliers; METR further shows that expected speedup can coexist with measured slowdown (developers forecasted faster completion yet were about 19% slower with AI allowed) [7]. Our workshop median (about $+100\%$ $\sum$ SP volume) is several times larger than those published experimental point estimates (e.g., $\sim$26% completed-task lift and $\sim$21% shorter task time in [3], [4]); constructs differ, so the comparison is a magnitude check rather than an effect-size meta-analysis, and the gap is itself a *falsifier* if later `metric_platform` rows do not converge toward a more moderate band. Keeping Throughput volume Δ and CT speed Δ as separate constructs (Section IV-C) is part of the same anti-myth discipline: two promissory dimensions must not be collapsed into one "$\approx 2\times$ everywhere" headline.

Modest maturity is news, not failure. In a $\sim$6-week pilot, clearing L1 shortfalls and lifting culture/process domains by fractions of a level is a plausible capability trajectory. Overclaiming "transformation complete" from Table VII would be as dishonest as suppressing the table because deltas look small. Readers should treat "limited movement in six weeks" as evidence about relative timescales of capability vs. promissory outcome signals.

Post-Explore measurement boundaries. Using the stage names introduced above: Explore dual-lens cells in this paper must not be extrapolated as Consolidation or Scale ROI—they remain Explore-window `process_maturity` and `metric_estimate` under facilitator presence. After consultants leave, maturity re-assessment done as client self-score would lack an independent assessor—a credibility gap we treat as a hard reporting boundary rather than a hidden After score. Process telemetry and board mapping between SDD changes and incumbent Agile practice are not yet `metric_platform`. We intentionally omit client-operated

TABLE VIII
CROSS-PAIR AGGREGATES OF IMPROVE Δ (ORG-A WORKSHOP). ALL ROWS: METRIC_ESTIMATE. CT = CYCLE TIME. SOURCE: EFFECTIVENESS_AGGREGATE.CSV.

Metric	Median	IQR	p25	p75	Min–Max
$\sum$SP improve Δ	1.017544	0.039384	1	1.039384	0.76–1.472727
CT1 speed Δ @P75	1	0.1875	1	1.1875	0.772727–1.714286
CT2 speed Δ @P75	1.018519	0.616402	0.75	1.366402	0.666667–1.619048

TABLE IX
PAIR SUMMARY (WORKSHOP): THROUGHPUT AND CYCLE TIME (CT) P75. ALL ROWS: METRIC_ESTIMATE. SOURCE: EFFECTIVENESS_PAIR_SUMMARY.CSV.

Pair	TP_b	TP_a	TP Δ	$CT1_b$	$CT1_a$	$CT1_\uparrow$	$CT2_b$	$CT2_a$	$CT2_\uparrow$
Pair-1	49	100	1.040816	3.75	1.875	2	5.5	2.1	2.619048
Pair-2	57	116	1.035088	3	1.5	2	5	3	1.666667
Pair-3	55	136	1.472727	4.75	1.75	2.714286	6.5	2.625	2.47619
Pair-4	45	90	1	4.875	2.75	1.772727	6.875	3.375	2.037037
Pair-5	50	88	0.76	4.5	2	2.25	5	2.5	2
Pair-6	50	100	1	4	2	2	5	3	1.666667

TP_b/TP_a: $\sum$ SP before/after. TP Δ: improve fraction $(\mathrm{TP}_a - \mathrm{TP}_b)/\mathrm{TP}_b$. CT$*_b$/$a$: Cycle Time P75 durations (CT1=code; CT2=delivery). CT$*_\uparrow$: speedup $= \mathrm{CT}_b/\mathrm{CT}_a$. Aggregate Table VIII uses speed-improve Δ = speedup-1; e.g., Pair-1 $\mathrm{CT1}_\uparrow = 2 \Rightarrow$ speed $\Delta = 1$.

Consolidation process telemetry from Results (owned by a companion stage-gated path paper); any such counts would remain `process_maturity` at best, never ROI. Part of the scheme's value is marking when an $N\times$ must *not* be reported. This manuscript stays on Explore dual-lens honesty and the empty After `metric_platform` cell.

VI. THREATS TO VALIDITY

We keep Results and Threats symmetric: every strong quantitative claim has a matching boundary here.

Construct validity. Story points are intended to capture inherent demand complexity independent of method; freezing SP supports separability but does not prove complexity invariance under AI4SE—spec-driven delivery may change how work *feels* complex even when points are frozen. Workshop $\sum$ SP and P75 CT operationalize delivery throughput and cycle time aligned to the metrics spec, yet **both Before and After** remain estimates under the fixed PBI list—not observed platform histories for either pole. The Before→After gap is therefore vulnerable to an "imagination gap": shared optimism about the new method relative to an also-imagined traditional baseline, not only to a measured baseline. Maturity L1–L5 rubrics and domain means are process constructs; they do not measure product quality or business outcome directly. Domain means average three ordinal levels; treat them as convenience summaries, not interval measurements. Choosing CT1/CT2 endpoints (last commit; QA testing completed) rather than production release reduces non-dev noise but also limits claims about lead time to customers. SPACE satisfaction, DORA reliability, and a dedicated quality before/after analysis are omitted by design (Sections II and III-C): retained client quality gates are a precondition for efficiency reporting, not evidence published in this paper.

Internal validity. Expert pairs may share optimistic (or conservative) bias on *both* Before and After; median/IQR reduce single-pair extremes but not shared optimism across poles. Facilitator framing, fixed conditions, and high-consensus After prompting shape estimates; this is not a blind controlled experiment. Holding Product Backlog / team / iteration window constant removes those three drifts; it does *not* establish that Before→After differences are solely caused by the method pack. We did not triangulate workshop Before against a matched platform window on the same PBIs in this phase—a deliberate comparability choice that increases estimate-to-estimate risk. Before/after maturity used the same facilitator-guided instrument; learning-to-the-test and social desirability may inflate small lifts. Blinded reassessment was *not* attempted in this pilot; we plan a next-phase repeat with a blind assessor (Section VIII). In a later Consolidation window with consultants off-site, self-scored maturity without an independent assessor would further weaken credibility; we therefore publish no Consolidation maturity After scores here. Other confounders (fatigue, concurrent org change, tool availability) remain. Independence of pairs reduces mid-workshop anchoring but does not eliminate shared prior beliefs about the method pack.

External validity. Single industrial case (Org-A); one $\sim$6-week enablement pilot in one software organization—not a multi-site sample. After estimates assume an After prompt framed as high-consensus SDD adoption; they do not generalize to cold-start adoption. Pair estimates under a fixed Product Backlog do not license portfolio- or enterprise-level productivity claims. Workshop grain is not organization grain.

Conclusion validity. All workshop productivity rows are `metric_estimate`; treating them as `metric_platform` ROI would be invalid. Modest maturity Δ neither proves "no effect" nor "transformation complete." With $n = 6$ pairs, median/IQR are descriptive summaries, not grounds for industry-wide multiplier laws or strong inferential claims.

Explore dual-lens numbers must not be retold as Consolidation or Scale ROI; Consolidation process telemetry, if any, stays `process_maturity` until `metric_platform` definitions and board mapping close.

Mitigations used: signal-type labels; fixed conditions + frozen SP; paired Δ then cross-pair median/IQR; promissory-expectation wording; dual-lens dashboard rather than a single multiplier; public derived CSVs so every table cell is auditable.

VII. Related Work

Methodological positioning. We follow software-engineering case-study reporting guidance on context, chain of evidence, and threats [8], while treating the dual-track scheme as a designed reporting artifact [9] illustrated in one engagement. We do not claim multi-case theoretical saturation under case-study reporting guidelines [8], researcher-led action-research ownership of Org-A change, or a completed design-science evaluate cycle with field-measured After outcomes.

Measuring developer productivity. Forsgren et al. argue that developer productivity is multi-dimensional and that single metrics mislead [1]. DORA research popularizes system-level delivery metrics (lead time for changes, deployment frequency, change fail rate, time to restore) [6]. Our dual-track protocol is an industrial operationalization of that multi-metric caution for AI4SE assessment: it pairs process maturity with stratified delivery signals, refuses a multiplier claim or single ROI number as the only management artifact, and explicitly omits full SPACE/DORA coverage under disclosure limits (Section III-C).

Empirical effects of AI coding assistance. Evidence ranges from larger controlled-task speedups (about 55.8% faster completion in one experiment) [5], through moderate field gains ($\sim$26% completed tasks) [3] and an enterprise-engineer lab RCT ($\sim$21% shorter task time, wide CI) [4], to documented gaps between perception and measured effect [7]. We use these ranges only to bound folklore $N\times$ claims and to motivate honesty labels on promissory estimates; we do not equate Org-A workshop medians with published experimental effect sizes, nor do we treat controlled-task speedups as portfolio ROI.

Maturity models for AI-assisted SE. Tereci et al. propose AASD-OMM, an organizational maturity model and research agenda addressing governance gaps in AI-assisted software development [10]. Garousi et al. introduce AI4SE-MM for individual and organizational competency measurement [11]. Anderson's AI Codebase Maturity Model frames progression from assisted coding toward more autonomous systems via feedback-loop topology [12]. Those works target broader organizational or codebase maturity theory; our niche is narrower: a consulting-embedded dual-track *assessment* protocol that jointly publishes a program-local 6×18 maturity checklist and stratified delivery-metric signals with an explicit estimate-vs-measured promotion path and a public derived pack for every reported cell—not a new universal maturity model and not a multi-site causal evaluation.

Niche statement. We contribute a boundary-aware management method for AI4SE pilots under disclosure constraints, with falsifiers and anti-misquote rules for sponsors, not a new causal effect size.

VIII. Future Work

This paper closes a phase-one enablement illustration with an empty `metric_platform` cell by design. Several follow-on tracks are required before any honest scale-out narrative.

Platform-measured iteration corpus. Continuously collect Throughput and CT1/CT2 from the requirements management platform across more real team iterations (matched definitions to the public metrics pack), so workshop `metric_estimate` medians can be validated or refuted rather than retold as ROI. Expand the published metric set beyond the workshop primary pair as instrumentation matures (e.g., additional workflow endpoints and team-grain aggregates), still under explicit signal-type labels.

Quality monitoring beside efficiency. This phase prioritized Throughput/CT because the sponsor ask was efficiency / productivity proof under Org-A's existing quality gates (Section II); those gates stayed as governance preconditions and were not instrumented as a reported parallel track here. Future work should elevate the same client-defined red-line indicators—e.g., defect escape rate and defect density—into a labeled quality track beside Throughput/CT, with the same estimate-vs-`metric_platform` discipline rather than a slogan-style quality multiplier. Scale-out claims for AI4SE change should then require non-degrading (or improving) quality under those definitions, not efficiency slogans alone.

Spec-driven development (SDD) metrics. Prioritize instrumentation for the spec-driven method pack itself: how specs/recipes are authored and reused, how Changes fold back to stories, and how SDD-specific workflow states relate to CT1/CT2 and completion under the same Product Backlog discipline. Those SDD metrics should enter the dual dashboard as labeled rows (estimate vs. `metric_platform`), not as a separate mythology.

Maturity gate for scale-out. Use repeated 6×18 maturity assessment as an explicit scale gate: prioritize teams that sustain domain/capability levels at L3+ (Engineered and above) for broader rollout, rather than copying promissory workshop Δs to cold-start teams. Combine that maturity gate with the estimate$\rightarrow$measured and quality tracks so AI4SE scale-out is admitted only when capability, delivery signals, and quality jointly meet an agreed expected band—not when a single multiplier looks attractive.

Assessor independence. Repeat maturity assessment with a blind (or at least independent) assessor to mitigate learning-to-the-test and social-desirability inflation acknowledged in Section VI.

For practitioners (organizational next step). Who in the sponsor circle owns the estimate$\rightarrow$measured promotion path; who co-signs signal-type labels on executive slides; and what governance pressure currently *punishes* distinguishing estimate

from measured—versus rewarding a single $N\times$ slogan—are Org-A-class questions that sit beside instrumentation. Without named ownership and incentives that protect honesty labels, `metric_platform` rows will not close the loop even when the platform can produce them.

None of these tracks are claimed as completed for Org-A in this manuscript; they define the honest next research and engagement agenda.

IX. Conclusion

We asked how AI4SE enablement improvement can be assessed jointly from delivery-result metrics and process-capability maturity (RQ1), and how dual-track assessment can be reported honestly when delivery metrics remain expert estimates rather than platform-measured values (RQ2). At Org-A, a $\sim$6-week pilot produced modest `process_maturity` lifts (domain Δ from $+0.33$ to $+1$) alongside `metric_estimate` Throughput/CT medians of roughly $+102\%$ $\sum$ SP improve and $\approx$ 1 CT speed-improve Δ—explicitly promissory, estimate-to-estimate, not `metric_platform` ROI. The case shows why process capability and delivery-result metrics should be reported jointly, and why a multiplier claim cannot stand in for either track. The dual-track dashboard, signal-type taxonomy, estimate$\rightarrow$measured promotion path (not yet closed here), falsifiers, and public derived pack are the transferable artifacts. Future work (Section VIII) centers on `metric_platform` iteration data, quality monitoring, SDD-specific metrics, and an L3+ maturity gate before scale-out. We do not extrapolate Org-A to industry-wide multipliers: multi-dimensional productivity guidance cautions against single-metric slogans [1], and published experiments already show moderate or even negative measured effects under some conditions [3], [7].

Acknowledgment

We thank Org-A engineering participants for enablement collaboration and executive readout discussion. We also thank Di Qin and Kun Wei (Inspire Group) for constructive peer feedback on an earlier draft of this manuscript.

Data Availability

Anonymized derived tables, metric definitions, workshop protocol text, and the maturity scoring guide accompanying this paper are released at https://github.com/ai4se-works/artifacts/tree/main/manuscripts/ai4se-measurement-maturity/data (CC-BY-4.0). Numbers in Tables VII–IX reproduce from `derived/*.csv` in that pack. Raw client requirements-platform extracts, identifiable workshop records, and named organizational materials are withheld under confidentiality constraints.

References

[1] N. Forsgren, M.-A. Storey, C. Maddila, T. Zimmermann, B. Houck, and J. Butler, "The SPACE of developer productivity," *Communications of the ACM*, vol. 64, no. 6, pp. 46–53, 2021, also appeared in ACM Queue, DOI 10.1145/3454122.3454124.

[2] L. Claxton, "The AI-native SDLC playbook," Anthropic Claude blog, aug 2026, online industrial playbook; author byline verified 2026-09-27. States that when build accelerates, plan, review/test, and deploy can remain human-speed bottlenecks. [Online]. Available: https://claude.com/blog/the-ai-native-sdlc-playbook

[3] K. Z. Cui, M. Demirer, S. Jaffe, L. Musolff, S. Peng, and T. Salz, "The effects of generative AI on high-skilled work: Evidence from three field experiments with software developers," *Management Science*, 2026, published online 27 Feb 2026; Crossref rechecked 2026-09-27 (volume/pages not yet assigned). Abstract reports 26.08% increase in completed tasks across three field experiments (SE 10.3%). Earlier draft circulated as MIT/SSRN preprint.

[4] E. Paradis, K. Grey, Q. Madison, D. Nam, A. Macvean, V. Meimand, N. Zhang, B. Ferrari-Church, and S. Chandra, "How much does AI impact development speed? an enterprise-based randomized controlled trial," 2024, preprint; abstract reports $\sim$21% shorter task time (wide CI); arXiv API rechecked 2026-09-27. [Online]. Available: https://arxiv.org/abs/2410.12944

[5] S. Peng, E. Kalliamvakou, P. Cihon, and M. Demirer, "The impact of AI on developer productivity: Evidence from GitHub Copilot," 2023, preprint; abstract reports 55.8% faster task completion; arXiv API rechecked 2026-09-27. [Online]. Available: https://arxiv.org/abs/2302.06590

[6] N. Forsgren, J. Humble, and G. Kim, *Accelerate: The Science of Lean Software and DevOps*. IT Revolution Press, 2018, dORA delivery metrics sourcebook; verify edition details at camera-ready.

[7] J. Becker, N. Rush, E. Barnes, and D. Rein, "Measuring the impact of early-2025 AI on experienced open-source developer productivity," 2025, preprint; abstract reports developers forecasted faster completion yet were about 19% slower with AI allowed; arXiv API rechecked 2026-09-27. [Online]. Available: https://arxiv.org/abs/2507.09089

[8] P. Runeson and M. Höst, "Guidelines for conducting and reporting case study research in software engineering," *Empirical Software Engineering*, vol. 14, no. 2, pp. 131–164, 2009, crossref verified 2026-08-30; online 2008-12-19.

[9] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," *MIS Quarterly*, vol. 28, no. 1, pp. 75–106, 2004, crossref verified 2026-08-30.

[10] S. Tereci, E. Gökalp, and A. Dikici, "Toward a maturity model for AI-assisted software development: Conceptual framework and research agenda," in *2026 5th International Informatics and Software Engineering Conference (IISEC)*, 2026, pp. 656–661, crossref verified 2026-08-30; proposes AASD-OMM (governance-oriented org maturity for AI-assisted SD).

[11] V. Garousi, A. B. Keleş, and S. Değirmenci, "Measuring individual and organizational competency in AI-driven software engineering: The AI4SE-MM maturity model," SSRN preprint, 2025, crossref DOI verified 2026-08-30; preprint / posted-content.

[12] A. Anderson, "The AI codebase maturity model: From assisted coding to fully autonomous systems," 2026, preprint; arXiv API verified 2026-08-30. [Online]. Available: https://arxiv.org/abs/2604.09388